



CAUSAL MAPS ARE KNOWLEDGE GRAPHS, BUT WITH WINGS

📅 26 Aug 2025

What is a Knowledge Graph? 🧠

- A knowledge graph is like a **giant mind map for a computer**. It stores information not as text in a document, but as a network of interconnected facts.
- It's built from two main things: **entities** (the "nodes," representing real-world objects, people, or concepts like "Paris" or "Photosynthesis") and **relationships** (the "edges," describing how these entities are connected, like "is the capital of" or "is a process in").
- A single fact has three parts: **(Subject) --- [Relationship] --> (Object)**. For example: (Marie Curie) --- [discovered] --> (Radium).
- Why are knowledge graphs specially useful in the age of AI? 💡
- **They create structure from chaos.** AI can read through millions of pages of unstructured text (like news articles or scientific papers) and pull out these factual triplets. This turns a messy sea of words into an organized, queryable database of knowledge.
- **They enable smarter searching and reasoning.** Instead of just searching for keywords, you can ask complex questions that require understanding the relationships between things. For example, "Which scientists who won a Nobel Prize also discovered an element?" A computer can navigate the graph's connections to find the answer.
- **They provide essential context.** A knowledge graph helps an AI understand that "Apple" in a tech article is a company linked to "Steve Jobs," not the fruit. By looking at its connections, the AI gets the right context, which is crucial for accurate understanding and analysis.

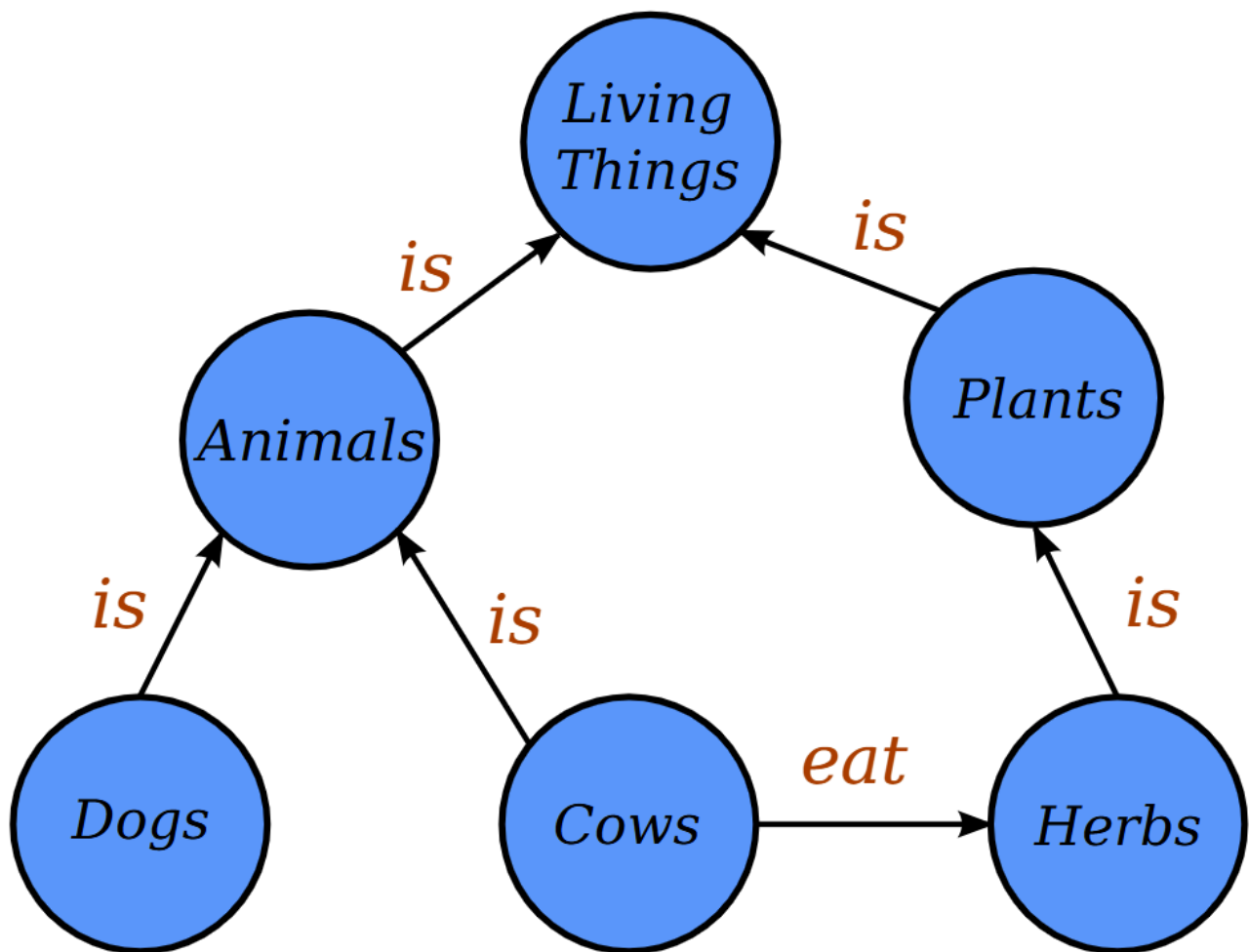


Image from Wikipedia by Jayarathina - Own work, CC BY-SA 4.0.

Why are Knowledge Graphs (KGs) so useful?

A major benefit of KGs is we can then apply network logic like transitivity rules to answer meaningful questions. For example, if the relation is "works in the same company as", then if we know

- **A is related to B**
- and
- **B is related to C,**
- then we can conclude
- **A is related to C (A is in the same company as C).**

Challenges with general-purpose knowledge graphs

The trick in **constructing** knowledge graphs is to know what relationship(s) to look for. "belongs to?" "is capital of?" "challenges/undermines?" This can be very difficult to decide. on the fly.

Causal mapping gives knowledge graphs wings



